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A Biopsychosocial Model for Understanding Training Load, Fatigue, and Musculoskeletal Sport Injury in University Athletes: A Scoping Review

Zachary J. McClean,[1,2] Kati Pasanen,[1,2] Victor Lun,[3] Jonathan Charest,[1,4,5] Walter Herzog,[1] Penny Werthner,[1] Amanda Black,[1] Reid Vander Vleuten,[1,2] Elise Lacoste,[1] and Matthew J. Jordan[1,2,3]

[1]Faculty of Kinesiology, University of Calgary, Calgary, Alberta, Canada; [2]Integrative Neuromuscular Sport Performance Lab, Faculty of Kinesiology, University of Calgary, Calgary, Alberta, Canada; [3]Sport Medicine Center, University of Calgary, Calgary, Alberta, Canada; [4]École de Psychologie, Université Laval, Québec, Quebec, Canada; and [5]Center for Sleep & Human Performance, Calgary, Alberta, Canada

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Abstract

McClean, ZJ, Pasanen, K, Lun, V, Charest, J, Herzog, W, Werthner, P, Black, A, Vleuten, RV, Lacoste, E, and Jordan, MJ. A biopsychosocial model for understanding training load, fatigue, and musculoskeletal sport injury in university athletes: A scoping review. *J Strength Cond Res* 38(6): 1177–1188, 2024—The impact of musculoskeletal (MSK) injury on athlete health and performance has been studied extensively in youth sport and elite sport. Current research examining the relationship between training load, injury, and fatigue in university athletes is sparse. Furthermore, a range of contextual factors that influence the training load-fatigue-injury relationship exist, necessitating an integrative biopsychosocial model to address primary and secondary injury prevention research. The objectives of this review were (a) to review the scientific literature examining the relationship between training load, fatigue, and MSK injury in university athletes and (b) to use this review in conjunction with a transdisciplinary research team to identify biopsychosocial factors that influence MSK injury and develop an updated, holistic biopsychosocial model to inform injury prevention research and practice in university sport. Ten articles were identified for inclusion in this review. Key findings were an absence of injury surveillance methodology and contextual factors that can influence the training load-fatigue-MSK injury relationship. We highlight the inclusion of academic load, social load, and mental health load as key variables contributing to a multifactorial, gendered environmental, scientific inquiry on sport injury and reinjury in university sport. An integrative biopsychosocial model for MSK injury in university sport is presented that can be used to study the biological, psychological, and social factors that modulate injury and reinjury risk in university athletes. Finally, we provide an example of how causal inference can be used to maximize the utility of longitudinally collected observational data that is characteristic of sport performance research in university sport.

Key Words: musculoskeletal injury, university sport, sport injury etiology, allostatic load

Introduction

The impact of musculoskeletal (MSK) injury on athlete health and performance has been studied extensively in youth sport and elite sport (15,29), including in university and collegiate athletes in the United States (56,58). However, current research examining the interrelationship between training load, fatigue, and injury that accounts for contextual factors specific to the university environment (e.g., academic load, social factors, sleep deprivation) is sparse (72). Sport injury is often modeled as a dynamic and recursive system (71,73) in which a previous injury is the leading predictor of a future injury (94), along with external risk factors (e.g., factors external to the athlete arising from the uncontrolled and dynamic sport environment) and internal risk factors (factors internal to the athlete; e.g., psychological stress, MSK tissue tolerance, or genetics) (73). These integrated models are crucial for informing primary and secondary injury prevention research (73) and are updated to account for emerging research and contextual factors that influence injury etiology (113). Key developments to the original dynamic recursive injury model (73) include a greater focus on the interrelationship between training load, fatigue, and injury and the importance of social factors such as the influence of gender on sport injury (82,113).

Recent research has also highlighted new approaches and statistical methods that can maximize the utility of observational data by quantifying causal relationships between various injury factors (54). For example, training load monitoring contributes to observationally collected longitudinal data in athlete populations. The term training load has been widely used by practitioners and researchers to quantify the dose-response relationship observed in training including the stress incurred from a training session that is associated with psychophysiological adaptations (33). A classification framework differentiates between the measurement of the external work done by the athlete known as the external load (e.g., total distance ran, total amount of weight lifted, ground reaction forces, training duration) and the athlete's psychophysiological response to training referred to as the internal load (e.g., perceived exertion, heart rate) (33). Training load is a key variable in the etiology of sport injury as

Address correspondence to Zachary McClean, zachary.mcclean@ucalgary.ca.

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the gradual application of training load increases the structural tolerance of MSK tissues (e.g., ligament, tendon, bone, and muscle) in a time-dependent manner to build resilience against injury and to support progressively higher training loads with time (116). However, in combination with biomechanical movement strategies, poorly timed or large shifts in training load are thought to contribute to an increased risk of acute and overuse MSK injury (79), along with an imbalance between training load, fatigue, and recovery (113).

Despite a broad body of scientific research aimed at primary and secondary injury prevention in athletes of all age groups and performance levels, no integrated, multifactorial, sport-specific, biopsycho-social model exists for sport injury and reinjury in university sport. Not only can such a model assist researchers in the development of context-specific solutions to address the burden of sport injury in this population but also it can facilitate a complex-systems approach to address the multifactorial nature of sport injury (9,32). For example, in the context of university sport, entry into the university setting typically occurs at the transition from adolescence to early adulthood, a phase marked by great biological variability, social pressure, and new academic challenges; evidence suggests that time- and gender-modulated factors influence a university student's capacity to cope with stressors specific to the university environment (54). Taken together, a holistic sport injury model that accounts for these factors along with the training load incurred by student athletes and the associated fatigue response can facilitate new approaches to address the multifactorial nature of sport injury and maximize the utility of data arising from university-based athlete monitoring programs (9,32).

Given the dearth of scientific evidence examining the interrelationship between training load, fatigue, and injury that accounts for contextual factors specific to the university athlete, an updated integrative sport injury etiology model is warranted to support research and practice in the context of primary, secondary, and tertiary injury prevention in this demographic. A key first step toward generating a new model is to map the body of the scientific literature on this topic. Therefore, the objectives of this review were to (a) review the scientific literature that has investigated the training load-fatigue-injury relationship in university athletes specific to MSK sport injuries and (b) use this review along with a transdisciplinary research team to identify biopsychosocial factors that influence the training load-fatigue-injury relationship to develop an updated, holistic biopsychosocial injury model specific to the university athlete.

Methods

Experimental Approach to the Problem

In adherence to the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews, a scoping review following the Population, Intervention/Exposure, Comparison, and Outcome framework was implemented to examine the training load-fatigue-MSK injury relationship. This search strategy was undertaken using subject headings and keywords, both explicitly and in combination, along with associated synonyms for 4 main concepts in the Embase, MEDLINE, PubMed, and SPORTDiscus databases in December 2022. These databases were searched for terms related to (a) university athletes (Population), (b) sport MSK injury (Outcome), (c) training load (Exposure), and (d) fatigue (both from performance and perceived perspectives (26) [Exposure]; Figure 1; see Figure A1, Supplemental Digital Content 1, http://links.lww.com/JSCR/A485). Only peer-reviewed articles written in English and published from 2002 to the time of this review were included for further review. Duplicates were removed, and for each remaining study, the title and abstract were screened independently by 2 reviewers (Z.M., M.J.) to identify relevant articles. In all cases, the 2 reviewers agreed on the relevance of explored articles.

When a relevant article was identified, the full text was then screened to ensure compliance with the inclusion criteria. The inclusion criteria for this review stated that the article must (1) present original research conducted on university athletes, (2) be

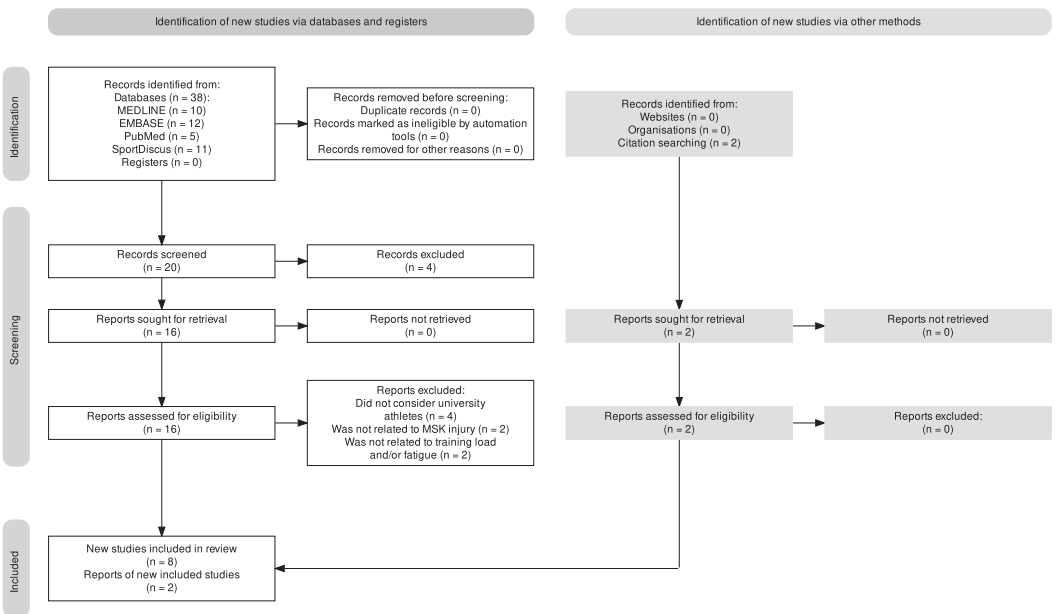

Figure 1. Schematic figure of the identification of articles for the present review. Created using PRISMA2020: An R package and Shiny App for producing PRISMA 2020 compliant flow diagrams (45).

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peer reviewed, and (3) consider the influence of training load (i.e., estimation of the stress induced by a given training session (33)) and fatigue on sport MSK injury (i.e., a MSK injury occurring in athletes as a result of their sport or sport-related training). Fatigue was classified as a decline in an objective measure of performance (i.e., performance fatigability) or a change in sensations that modulate the integrity of the exerciser (i.e., perceived fatigability) (26). Articles that examined *training load* and *fatigue* in the context of MSK injury as independent constructs and in conjunction as interrelated constructs were considered for this review, along with articles that measured MSK injuries or that discussed a potential relationship with MSK sport injury. After removing duplicates, this search strategy resulted in the identification of 8 articles that fit within the inclusion criteria and therefore were suitable for this review (Figure 1). The reference lists of the selected articles were cross referenced and 2 additional articles were added, leading to 10 articles suitable for review. Figure 1 provides an overview of the identification, retrieval, and analysis of the relevant scientific articles. The Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies (75) was used to guide a methodological assessment for each study (Tables 1 and 2).

Consistent with the second objective of this study, experts from the domains of sport medicine, sleep medicine, sport psychology, clinical biomechanics, and sport science were consulted to build off the findings from the literature search and identify contextual factors relevant to the training load-fatigue-MSK injury relationship in university sport. The transdisciplinary research group explored the training load-fatigue-MSK injury literature and achieved consensus on additional biopsychosocial factors relevant to the university sport setting. The group's operational process constituted semistructured, consensus-building, iterative meetings, whereby a broad set of contextual factors were proposed by the domain experts and then reduced to the final list of 8 contextual factors achieved by group consensus. These factors were deemed to be of highest relevance to a model aimed at describing the training load-fatigue-MSK injury relationship using a biopsychosocial approach. After achieving consensus on the 8 contextual factors, the domain experts synthesized relevant research and contributed to the development of the integrative training load-fatigue-MSK injury model presented in Figure 2. Simplified model frameworks were initially developed and iterated upon through group discussion to achieve the final model presented in this review. This study was approved by the University of Calgary Conjoint Health Research Ethics Board (RED15-1094).

Results

A total of 10 articles were included in this review. Of the total number of selected articles, 9 studies focused on National Collegiate Athletic Association (NCAA) athletes in the United States, and 1 was conducted on Canadian university sport (U Sport) athletes. A total of 319 subjects were included across all the selected studies (mean ± *SD*; mean study sample size = 31.9 ± 18.9), of which 14.1% were female. Of note, 2 studies included only females, and no study considered both male and female athletes together. Two authors (Z.M. and M.J.) reviewed the selected articles and, based on each study's primary exposure variable(s), synthesized them into 2 major themes: (1) *the training load and MSK injury relationship* (articles: $n = 7$) that included articles with exposure variables related to internal or external training load; and (2) *the fatigue and MSK injury relationship* (articles: $n = 3$) that included articles with exposure variables related to perceived fatigability and performance fatigability. In cases where the reviewing authors disagreed on an article categorization, the article was discussed until a consensus was reached. A summary of the methodologies and main findings of these articles for the 2 themes of this review are presented in Tables 1 and 2.

Regarding the second objective of this review, the transdisciplinary team identified 8 additional contextual factors not identified in the literature review that were determined to be important for a training load-fatigue-MSK injury model in university sport. These biopsychosocial factors included the following: (1) establishing the athlete baseline and appropriate injury surveillance methods to detect a change in this baseline with respect to MSK injuries; (2) elucidating injury mechanism(s) in a sport-, sex-, and gender-specific manner; (3) monitoring sleep and sleep-related factors; (4) academic load; (5) mental health load; (6) social load; (7) gendered environmental factors; and (8) the role of sport and institutional norms in the training load-fatigue-MSK injury relationship. These topics are discussed in Part 2 of the Discussion.

Discussion

The primary finding of this review was the relative paucity of the scientific literature examining the interrelationship between training load, fatigue, and MSK injury using a contextual approach specific to the university sport setting. Given the scale of university sport in North America, the abundance of the literature on training load and injury in other sport settings, and several epidemiological studies of MSK injury in university sport (72,89), the lack of scientific studies using a holistic approach to study the training load-fatigue-injury relationship in university athletes was unexpected. This is problematic considering recent injury rate estimates from both Canadian university sport (38) and the NCAA (1) suggests a high incidence of MSK injuries in the university athlete population. For instance, lower-extremity MSK injuries are common in high-risk sports, such as hockey, tackle football, and basketball, in both male and female athletes where the ankle and knee are most injured body parts (72,89). Considering that injuries to the ankle and knee are likely to be severe (>28-day time loss) and lead to secondary and tertiary consequences (e.g., reinjury, osteoarthritis) that impose a substantial burden on the university athlete, the university operated health care system, and on the health care system at large (25), it is critical that MSK injuries in this population are investigated from a holistic viewpoint.

The fact that our search strategy specific to the training load-fatigue-MSK injury relationship yielded few scientific publications demonstrates a knowledge gap in the context of university sport, especially in U Sport. This is notable due to the potential differences in cultural factors and resources between American and Canadian programs that could modulate the proposed training load-fatigue-MSK injury pathway in university athletes. Furthermore, although the relationship between training load, fatigue, and injury has been more thoroughly investigated in American university athletes (28,87), the breadth and depth of this literature neglected important methodological considerations that limited interpretation of the results (e.g., inadequate injury surveillance, not controlling for athlete sport exposure, a lack of sex- and gender-based analysis, limited consideration for performance level). Training load was also used as a proxy for MSK tissue load (i.e., an implicit assumption that the greater the training load, the greater the load on MSK tissues and therefore injury risk), a notion that has received criticism in the scientific literature (53). Furthermore, a holistic biopsychosocial analysis that included the physiological, psychological, and social stressors impacting the university athlete inside and outside of the

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Table 1

Summary of studies included in the training load and injury relationship theme of this review.*

First author	Journal (year)	Study population	Study design	Parameters investigated	Main findings and conclusions	Risk of bias (90)
Williams et al. (110)	Science and Medicine in Football (2019)	NCAA Division I women's soccer ($n = 25$)	Cross-sectional cohort	Global positioning system data, total distance, high metabolic power distance, speed exertion, heart rate exertion, average heart rate, energy expenditure	Soccer extra time added 20–25% increase in player training load Training loads during extra time were not different between players who did and those who did not play the entire regular time There was no evidence of decreased work rate (i.e., fatigue) during extra time	Good
Wilkerson et al. (109)	Journal of Strength and Conditioning Research (2016)	NCAA Division I men's football ($n = 45$)	Retrospective cohort (15 consecutive weeks)	IMU data, player inertial load, intraindividual inertial load variability, injuries resulting in at least 1 d of limited football participation	The number of days of injury avoidance before a first injury were more related to participation in >289 football plays than individual player ILV When adjusted for game exposure, a low degree of ILV, and high absolute inertial loads had a cumulative hazard across 15 wk	Good
Fields et al. (30)	International Journal of Exercise Science (2020)	NCAA Division I women's lacrosse ($n = 20$)	Prospective cohort (13 wk)	Heart rate variability, salivary samples (analyzed for free testosterone and cortisol), perceived measures (perceived recovery scale, overall fatigue scale)	No significant time effects from week 1 to 13 were found for any variable. However, heart rate variability, testosterone, and perceived recovery, cortisol, and perceived fatigue demonstrated a 12, 15, 18, −5 and −2% change from week 1 to 13, respectively	Good
Murray et al. (76)	Journal of Science and Medicine in Sport (2019)	NCAA Division I men's Football ($n = 63$)	Prospective cohort (15 wk)	IMU data collected during training and games, player load, within and between stride waveform variability measures, wellness questionnaires (areas and severity of soreness [1–10 scale], as well as fatigue, sleep quality, and overall muscle soreness measures [1–5 scale]), suspected hamstring, ankle, foot injuries (i.e., "flagged status")	Increased within and between stride waveform variability were associated with increased fatigue, soreness, and player load No association between stride variability and injury location was found	Good
Roy et al. (86)	Journal of Australian Strength and Conditioning (2020)	U Sports Men's Football ($n = 26$)	Prospective cohort (85 d, 2017 U Sports football season)	Training load (calculated as the multiple of session-RPE [0–10] and session-duration [mins]), acute (7 d) to chronic (28 d) workload ratio	Acute training load increased during first 3 wk (training camp) Acute training load progressively decreased from days 21–85	Poor
Govus et al. (42)	International Journal of Sports Physiology and Performance (2018)	NCAA Division I men's Football ($n = 58$)	Retrospective cohort (8 wk)	IMU data (3 sessions/week), player load, posttraining subjective measures (session-RPE [0–10 scale]), pretraining subjective wellness (perceived muscle soreness, sleep, energy [1–5 Likert scale])	Increased perceived muscle soreness was related to a small decrease in session RPE Increased wellness *z*-score and perceived energy were both associated with increased player load Muscle soreness and sleep were not related to player load	Good

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Table 1

Summary of studies included in the training load and injury relationship theme of this review.* *(Continued)*

First author	Journal (year)	Study population	Study design	Parameters investigated	Main findings and conclusions	Risk of bias (90)
					Perceived energy and sleep were not related to session RPE	
Sampson et al. (87)	Journal of Strength and Conditioning Research (2019)	NCAA Division I men's Football (n = 42)	Prospective cohort (17 wk)	Injury occurrence (noncontact injuries requiring withdrawal from practice or game time), IMU data (all weekly training sessions and games included), player load, daily exponentially weighted moving average acute (7 d) to chronic (21 d) workload ratio, pretraining subjective wellness (perceived muscle soreness, sleep, energy [1–5 Likert scale])	Fifty-five noncontact injuries occurred (n = 42 lower-body injuries) Injury risk was higher when ACWR was 2*SD*s above the mean or 2*SD* below the mean No effect of injury likelihood with reported increased or decreased subjective wellness, sleep, energy, or soreness Lower wellness, energy, and higher muscle soreness were reported after training sessions that evoked ACWR >2*SD* above the mean response The greatest risk of injury (within 3 d) occurred when ACWR was >2*SD* above the mean, *better* perceived wellness, and muscle soreness collectively occurred	Good

*NCAA = National Collegiate Athletic Association; IMU = inertial measurement unit; RPE = rating of perceived exertion; ACWR = acute-to-chronic workload ratio; ILV = inertial load variability.

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sport environment was not provided. Consequently, to best synthesize the findings of our search strategy, articles were grouped into a *training load-MSK injury* theme and a *fatigue-MSK injury* theme.

Training load monitoring is recommended to identify overreaching, underrecovery, or undertraining that is thought to increase injury risk (53,113), and recently an increased emphasis has been placed on the importance of daily athlete monitoring (37). In the present review, a range of variables were monitored across studies including internal training load (heart rate, within and between limb stride variability measures, and sessional rating of perceived exertion [sRPE]), external training load (total distance ran, high intensity running distance, and acceleration load), and introspective measures (muscle soreness, energy levels, and sleep quality). However, there was a notable absence of studies that examined training load and injury concurrently. For example, a single Canadian U Sport study found in our literature search on tackle football players monitored training load using the sRPE method over an 85-day tackle football season and found that sRPE fluctuated throughout the season with the highest values occurring during training camp and the lowest values toward the end of the football season (86). However, accepted injury surveillance methodologies were not strictly followed along with no measurement of the external training load, which limited an in-depth examination of the relationship between training load and injury. The combined use of external training load, internal training load, and introspective wellness measures have been previously implemented to gain insight regarding fatigue (44,99), training load (42,86), and factors that promote recovery such as sleep (107). These variables are thought to be fundamental for athlete monitoring programs (48,90).

Williams et al. (110) used global positioning system to quantify total running distance, high-intensity running distance, and the number of high speed running bouts during NCAA women's soccer games with and without extra time and found that in extra time, players demonstrated a capacity to maintain high work rates despite the presence of a greater than 20% increase in external load. However, injury surveillance was also not conducted in this exploration, and the psychological or social factors that may contribute to sport performance and injuries were not measured. Finally, a study using inertial measurement unit (IMU) technology to monitor external load in NCAA men's tackle football players during a training session, in conjunction with pretraining subjective wellness parameters (i.e., Likert scales assessing muscle soreness, sleep quality, and energy level) and posttraining sRPE (42), found an association between self-reported muscle soreness and sRPE (42). However, these introspective measures should be interpreted with caution because the subjective parameters assessed were unvalidated and may be prone to bias because of the utilization of Likert scales. Nevertheless, this study does highlight the likely interplay between psychophysiological factors and an advantage of coupling variables to inform sport injury research models in a university sport setting.

Two studies examined training load with respect to injury factors (87,109). Sampson et al. (87) examined the relationship between training load and noncontact injuries causing missed practices and games in tackle football players. Forty-two injuries were recorded, and a ratio of the 7-day training load to the 21-day training load (acute:chronic load ratio) was associated with increased injury risk. Notably, the greatest risk of injury occurred with the acute:chronic load ratio of >2 *SD*s above the individual mean, when perceived

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Table 2

Summary of studies included in the fatigue and injury relationship theme of this review.*

First author	Journal (year)	Study population	Study design	Parameters investigated	Main findings and conclusions	Risk of bias (90)
Watson et al. (107)	Orthopaedic Journal of Sports Medicine (2020)	NCAA Division I male basketball athletes ($n = 19$)	Prospective cohort	Perceived fatigue, mood, soreness, stress, and sleep quality (0–5), sleep duration (min), subjective intensity of physical activity (1–10), injuries (an acute or chronic injury resulting in time missed from typical training or game participation)	Sleep duration ($P < 0.001$) and soreness ($P = 0.024$) were significant predictors of injury	Good
Grantham et al. (44)	Orthopaedic journal of Sports Medicine (2014)	NCAA Division I male baseball pitchers ($n = 11$)	Prospective cohort	Number of pitches thrown, time between pitches, kinematic pitch analysis variables, perceived fatigue (0–10, implemented post game)	Increased hip lean, hip flexion, elbow height, and shoulder tilt occurred throughout throwing motion in innings lasting longer than 15 pitches Season pitch count was correlated with increased shoulder external rotation ($r = 0.42$), and altered shoulder mechanics at ball release Elbow flexion decreased with greater pitch counts over the course of a season Subjective fatigue was correlated with reduced knee flexion ($r = 0.51$) and glove height ($r = 0.46$) during throw	Good
Escamilla et al. (27)	The American Journal of Sports Medicine (2007)	NCAA Division I male baseball pitchers ($n = 10$)	Observational cohort (simulated game with 105–135 pitches)	Ball velocity (miles·h^{-1}), 3D kinematic variables	As a pitcher approaches subjective muscle fatigue, significantly less ball velocity occurred, and the trunk was significantly more vertical ($P < 0.05$) As a pitcher approaches muscle fatigue, shoulder and elbow torques and forces were not increased	Good

*NCAA = National Collegiate Athletic Association.

wellness increased, and when muscle soreness decreased. The authors postulate that this surprising finding might be explained by the phenomenon whereby higher-intensity efforts or riskier behaviours occur more frequently when athletes felt "better"; furthermore, it is possible that athletes who have higher perceived wellness may be less likely to seek out athletic therapeutic interventions in comparison to those with lower wellness and higher muscle soreness (87). In addition, Wilkerson et al. (109) examined the relationship between external training load (measured with IMUs) and MSK injuries, resulting in at least one missed practice. When adjusted for game exposure, the combination of reduced variability in the external load and a high absolute external training load over a 15-week time period increased injury risk. Although limitations of the acute:chronic workload ratio have been presented (53,112), these studies offer insight into how shifts in training load may increase the risk of injury.

A potential constraint contributing to the absence of data on the training load-injury relationship in university sport may stem from the fact that training load was often used a proxy for MSK tissue load, which has been debated (36,53). There is no current, globally accepted, and valid standard for training load quantification, and given that the literature examining the relationship between training load and injury is in its infancy for university sport, it appears necessary to (1) include measures of internal and external training load (70); (2) to consider measuring MSK tissue structural tolerance through quantification of biomechanical load (103); and (3) to combine training load measures alongside introspective measures. Further to this point, Kalkhoven et al. (53), have argued that current training load measures are likely too distant from an inciting injury event for meaningful relationships to be drawn. Here, the inclusion of fatigue responses between training load and sport injury may assist in elucidating the underlying mechanisms associated with sport injury.

There was limited research examining the relationship between fatigue and injury in university sport, which may be attributable to a heterogenous definition of fatigue within the sport science community (26,60). Recently, it has been proposed that fatigue be assessed with respect to performance fatigability (i.e., a decline in an objective measure of performance due to fatigue) (18,46) and perceived fatigability (i.e., changes in sensation that modulate the integrity of the exerciser) (26,49). In applied sport science

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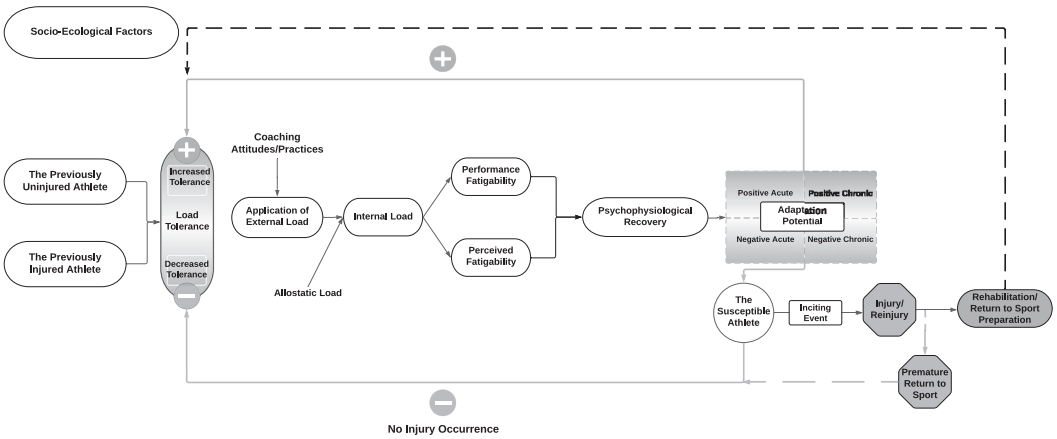

Figure 2. An integrative biopsychosocial model for sport injury in university sport. This model describes the proposed relationship between training load (e.g., game load, practice load, and training load), allostatic load (e.g., sleep load, academic load, mental health load, and social load), internal load (e.g., psychophysiological responses), performance fatigability (e.g., neuromuscular function, biomechanical function, and physiological function), and psychophysiological recovery on adaptation potential and MSK injury. In addition, this model considers the overarching influence of socio-ecological factors (e.g., gender, ethnicity, attitudes/beliefs towards training) on the aforementioned relationship. MSK, musculoskeletal.

literature, a causal relationship between fatigue and MSK injury has been proposed due to the fact that performance fatigability has been shown to exacerbate high risk biomechanical movement strategies (115). Furthermore, perceived and performance fatigability are also thought to contribute to reduced performance in university athletes with respect to the sport load and nonsport load (e.g., academic load, social load, mental health load) (83).

This review demonstrated inconsistencies between sport injury research in a university sport setting and accepted methodologies within fatigue science, which impaired conclusions on the training load-fatigue-MSK injury relationship. For instance, of the studies reviewed in this article, a single study examined athlete fatigue responses in conjunction with accepted injury surveillance methodologies (107). In a prospective cohort study, Division I basketball players (n = 19) were monitored using introspective measures and sleep duration. Sleep duration and muscle soreness were found to be significant predictors of injury risk (107), highlighting the relevance of monitoring sleep related factors, such as sleep duration, quality, and latency in university athletes.

In addition, our review highlighted several articles that indirectly considered the fatigue-MSK injury relationship by examining the effects of fatigue on sport biomechanics and common

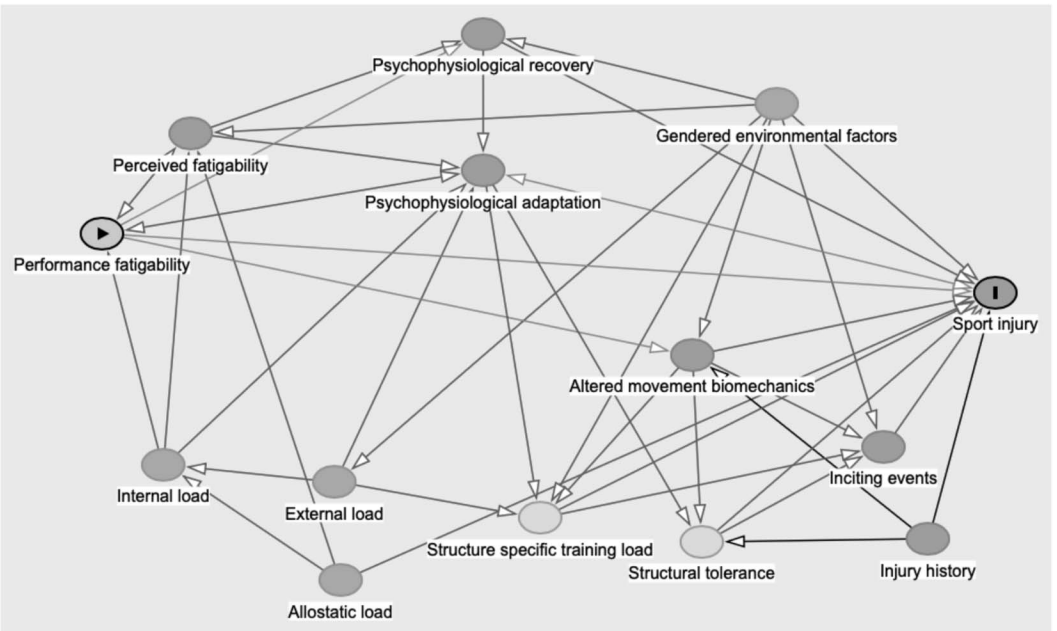

Figure 3. A directed acyclic graph derived from the an integrative biopsychosocial model for sport injury in university sport that examines the relationship between performance fatigability and sport injury in the context of university athletes. Green circle with inset triangle represents exposure variable; blue circle with inset line represents outcome variable; blue circle represents an ancestor of the outcome; pink circle represents an ancestor of both the outcome and exposure; grey circle represents an unobserved, latent variable; green connecting line represents causal pathway; pink connecting line represents biasing pathway; black connecting line represents neither causal nor biasing pathway.

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biomechanical injury mechanisms. For example, Escamilla et al. (27), examined the effects of perceived fatigability on throwing kinematics in NCAA male baseball pitchers demonstrating that fatigability was associated with reduced ball velocity and a more vertical trunk position during throwing. Adopting a similar methodology, Grantham et al. (44), assessed the effects of fatigability on throwing kinematics in NCAA male baseball pitchers, yet, showed fatigability-dependent changes in throwing mechanics different from Escamilla et al. (27). Previous studies highlighted that biomechanical alterations during a fatiguing task occur in an individualized manner (81,111); this might explain divergences in the results presented by Escamilla et al. (27), and Grantham et al. (44). Nevertheless, biomechanical movement patterns are thought to contribute to mechanisms of noncontact sport injuries such as anterior cruciate ligament(ACL) rupture (e.g., increased knee abduction moment (14)) and should be considered on an individual basis with statistical methods that can integrate covariates such as fatigue and training load. Examples of these methods include generalized additive models (52) and machine learning (91).

In summary, the research on the fatigue-injury relationship in university athletes was limited. Nevertheless, the notion that fatigue is a contextual factor that increases the risk of sport injury has been postulated (51,98), but conflicting research exists (8,11). Based on the research, fatigue monitoring should include assessments of performance fatigability (e.g., vertical jump kinetics and kinematics) (39) and perceived fatigability, along with elucidating how fatigue may increase high risk biomechanical movement strategies.

Building from the results of the review, a transdisciplinary team approach was used with experts from sport psychology, clinical sports medicine, clinical biomechanics, sleep science, and sport science to identify additional contextual factors related to the training load-fatigue-MSK relationship in university sport. Eight factors are discussed with respect to the biopsychosocial injury model presented in Figure 2.

Because of the possibility of divergent training and injury histories of university athletes, appropriate preseason screening that assesses biological factors (e.g., age, sex, neuromuscular function, muscular strength, and movement mechanics), psychological factors (e.g., psychological readiness for sport and academics, athlete beliefs), and social factors (e.g., gender, race, socioeconomic status, coaching attitudes/practices) to establish the *athlete baseline* is critical. Here, the athlete baseline can inform both scientific research and an operational strategy to support university medical staff in promoting athlete health and safety. Injury surveillance methodologies were inconsistent between studies in this review but it remains a crucial component of sport injury research (101). In addition, given the influence of previous injury on the risk for future injury, questionnaires like the Sport Fitness Index may help researchers quantify a university athlete's injury history at the time of the athlete baseline (108).

Although concussion injury mechanisms have been explored extensively in university sport (21,118), there is a paucity of studies that have examined the MSK injury mechanisms, particularly within Canadian university sport. Elucidating the injury mechanisms is key for developing appropriate injury prevention strategies (101). Outside of the university sport setting, injury mechanisms have been described using systematic video analysis (66,80) despite potential limitations of this methodology (114). In addition, the predominant MSK tissue-specific mechanisms of secondary sport injuries may be distinct from those of primary injuries. For example, athletes are twice as likely to injure their uninjured (contralateral) ACL after an ACL reconstruction on the ipsilateral limb. Nevertheless, describing sport-specific injury mechanisms from a multidimensional lens may provide greater insight into the inciting event and could be considered in a holistic, biopsychosocial sport injury model.

Injury prevention models have frequently been updated to reflect new evidence and contextual factors that may be implicated in the etiology of sport injury. For example, the transition from high school to university is characterized by biological, psychological, social, and academic development, which brings new challenges for university students (78), and potential interactions with injury risk factors. Given the potential for this period to impair mental health, sleep, and recovery (12,63), all of which have an impact on an athlete's adaptive potential and possibly their injury resilience, a biopsychosocial model for sport injury that is specific to university athletes must consider these covariates. Additional factors like the sport culture and the gender-disparity in sport medicine and training resources should also be considered in a multi-factorial sport injury model (82).

Sleep-related fatigue carries with it substantial health and performance implications (35), including a potential relationship with MSK injury (104) and sport concussion (84). This review showed that sleep duration and muscle soreness predicted injury in NCAA basketball players (107). Similar findings highlighting the importance of sleep quality and quantity in sport injury prevention have been reported in elite soccer players (92), and others have shown that sleep is not related to injury risk when athletes obtain sufficient sleep duration and quality (20). However, elite athletes (7) and university athletes (67) are at high risk for poor sleep duration and sleep quality. University athletes are thought to be at greater risk for the development of sleep disorders compared with the general student body (74). Lifestyle factors (e.g., sleep hygiene, part-time employment). psychological distress (e.g., combined academic and sport related stress, motivation loss), and the intense demands of the university sport schedule are thought to increase the risk of sleep disorders in university athletes (74). Further, when compared with their nonathlete counterparts, university athletes are less sedentary outside of sport-specific training times; this may provide evidence for a greater sleep need in this demographic (23). Despite the larger apparent sleep need, Driller et al. (23), demonstrate that university athletes tended to sleep less and displayed worse sleep patterns compared with a university student control cohort. Sleep is essential for recovery, neuromuscular performance, cognitive function, psychological wellness, and immune function, all of which are foundational to sport and academic performance, injury prevention, and general health and wellness in this group (13,102), supporting the inclusion of sleep monitoring in a biopsychosocial sport injury model. This includes the use of baseline sleep questionnaires in athlete monitoring to identify sleep disorders that are prevalent among university athletes (69,74,88).

Despite the similar academic expectations of university athletes and the general student body, the concurrent academic and athletic requirements is a unique challenge that increase stress compared with that of typical university students (4,65). For example, the number of hours required for sport-specific training, non-sport specific training, competition, travel, and general sport preparation is thought to increase fatigue, which can then interfere with an athlete's capacity to engage in academic tasks, and consequently increases their overall academic load (4). Further, some university athletes experience demanding sport travel schedules that disrupt academic attendance and study schedules leading to an accumulation of academic tasks upon returning from travel, creating

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intense microcycles of academic load that are uncommon in the general university population (4). Although research investigating the link between academic load and sport injury is sparse, Mann et al. (68) retrospectively analyzed injury occurrences (i.e., an injury that reduced practice participation) in relation to weeks with low academic load (i.e., training camp weeks that occurred before the semester), medium academic load (i.e., typical academic weeks), or high academic load (i.e., exam weeks) in NCAA Division I men's tackle football players. This investigation found that periods of high academic load were associated with increased risk of injury. Further, Davis et al. (19) found that academic stress was positively correlated with mental health dysfunction, sleep disturbances, and perceived fatigability in alpine skiers. Finally, poor mental health and sleep function, along with increased perceived fatigability led to greater injury and illness susceptibility in university athletes (47). Given the cyclic nature of the academic schedule that results in periods of intense academic demands, academic load should be considered in a biopsychosocial approach to athlete monitoring for the university athlete.

In recent years, there has been an increased awareness on the mental health of university students. The high school to university transition can present challenges related to identity formation, socioeconomic status, and the reconciliation of previous traumas; these factors, and others, can increase susceptibility for mental health impairments (96). University athletes remain sensitive to these issues, along with additional psychological challenges pertaining to the concurrent demands of academics and athletics. In addition to time management stress, university athletes experience a range of expectations and pressures from internal (e.g., pressure for individual accolades in academics and sport, the desire to achieve a professional sport career) and external (e.g., pressure to please coaches, family, and teammates) sources (96).

The mental health load may be high in the university athlete and can exacerbate the risk for anxiety and depression symptoms (55,100), with previous studies illustrating that in NCAA Division I university athletes, 21% reported symptoms of depression, 29% reported symptoms of generalized anxiety disorder, and of those who reported depression or anxiety symptoms, 49% reported symptoms of both depression and anxiety (61,64). Although these rates are lower than what Beiter et al. (6), reported in the general student body (40 and 33% of undergraduate students who experienced anxiety and depression, respectively), the underlying mechanisms associated with these mental health conditions are thought to be different between university athletes and the general student body (64). Further, it is of note that the risk of a mental health disorder has been frequently demonstrated to be increased with multiple exposures to traumatic brain injuries, which occur frequently in university sport (57). In an attempt to address these issues, mental health and support services have been frequently implemented for university athletes; yet, their effectiveness has been debated (75). Emerging evidence suggests that from a mental health perspective, 1 benefit of the university sport setting is the team-orientated social support network that supports enhanced athlete mental health (95). Importantly, sport injuries often lead to athletes feeling isolated from this critical social support network because of a reduction in team engagement through absence from activities such as practices, games, and travel (10,93). In the context of sport injury, although the mental health repercussions of injury and acute removal from team-orientated social support have been explored repeatedly with literature suggesting strategies for the inclusion of injured players in team events (e.g., practices, travel), to the best of our knowledge no study has explored mental health as a predictor of sport injury in university sport.

Social load has frequently been viewed as an important aspect of the transactional model of stress and coping where social support may be protective of the stress induced by academics, athletics, and injury (10,31). However, during the high school to university transition, the emotional and cognitive-social maturity of university athletes is variable and likely influences their capacity to cope with life factors such as social overload and social conflict (78). Therefore, although a comprehensive social support network may have great value for the university athlete, it is important to also consider that social challenges and pressures can induce stress that could modulate the short- and long-term university athlete experience and their response to injury. In a sport injury prevention model, the quantification of social load is complex. However, Alexe et al. (2), stated that a comprehensive assessment of perceived social support in university athletes can be achieved through multiple athlete self-reported measures using questionnaires such as Perceived Stress Scale (16) or the Multidimensional Scale of Perceived Social Support (117). Recently, Grandner et al. (43) used these self-reported measures in the investigation of mental health and sleep in NCAA athletes and demonstrated that reduced sleep quality and quantity was associated with increased stress, depression, and anxiety, as well as decreased social support. Consistent with the primary theme of this review, social load appears to be a key factor contributing to a holistic biopsychosocial model for sport injury in university sport, along with emphasizing the need for greater resource for psychosocial interventions to support the mental health of the university athlete.

When considering the biopsychosocial investigation of sport injury, emerging evidence has recommended a gender-based-analysis-plus approach whereby the entanglement of biological sex and gender (i.e., a social construct) are considered in the analysis of injury risk (3,82). In their review, Parsons et al. (82), noted that the current literature investigating ACL injury in sport frequently adopts a reductionistic approach that only investigates biological sex differences involved with injury mechanisms. In doing so, this approach has the potential to proliferate gender- and sex-bias in sport research (82). An alternative approach is to examine potential modifiable and non-modifiable risk factors internal to the athlete (i.e., sex-based factors such as skeletal muscle strength and body composition) and external to the athlete (i.e., gender-based factors such as access to sport, training and medical resources) (34,82). To this point, gendered environmental factors should be further studied in comprehensive methodologies of sport injury prevention, and included in a biopsychosocial injury model specific to university sport.

Finally, social factors may impact injury risk are the societal norms associated with sport and university institutions. Although the current literature has explored these concepts to some extent in traumatic brain injury (62,85), the primary sociological factors modulating university sport MSK injury risk have yet to be fully elucidated. Sport norms have been shown to influence athlete engagement in evidence-based injury prevention strategies in other settings (17), highlighting the inclusion of sport and institution norms to address social factors that may contribute to injury risk in university sport.

Based on this review, the work of Jeffries et al. (50), was expanded upon to develop a biopsychosocial model to assess the training load-fatigue-injury relationship in university athletes (Figure 2). Importantly, this model accounts for previous injury history and reinjury to account for persistent psychological (5,40), biomechanical (24,41,97), and physiological (106) deficits arising from an MSK injury, and the practice of seasonal athlete

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baseline testing to identify mental health conditions, along with contextual factors such as sleep, mental health, and academic loads that may modulate performance and injury risk. Collectively, these psycho-social factors contribute to the *allostatic load* that is thought to regulate stress adaptation in athletes (59).

Given that longitudinal athlete monitoring is an accepted practice in university sport, including for injured athletes with respect to secondary and tertiary injury prevention, high time resolution measurements, and appropriate statistical methods can be used to understand the training load-fatigue-MSK injury relationship. This includes methods such as machine learning and causal inference to maximize the value of longitudinally collected data (22,105). In Figure 3, we provide an example of how the larger sport injury model presented in Figure 2 can be used to inform a more focused research question using causal inference. Here, we constructed a directed acyclic graph (DAG) derived from the larger sport injury model presented in Figure 2 to study the relationship between performance fatigability and sport injury. Not only does the DAG provide the scaffolding for future statistical analysis but also it requires a transdisciplinary approach for establishing pathways of causal effects, mediators, and confounders that are determined at the outset of the scientific planning process. These approaches may support researchers in the university sport setting who aim to understand the relationship between training load, fatigue, and MSK injury using a biopsychosocial approach.

Practical Applications

Although incorporating the concepts discussed within this scoping review into research and applied sport science settings is likely to improve the understanding of the training load-fatigue-MSK injury relationship, the complexity of this problem cannot be overstated and implementation of a comprehensive research model like the one proposed here is not without potential challenges. For example, the model presented in this review is currently being employed in a related longitudinal research study by the authorship team, and addressing challenges related to feasibility and practicality has depended heavily on interdepartmental collaboration between research faculty, athletic therapy or sports medicine staff, strength and conditioning coaches, and athletic management at the study outset to identify relevant research questions and allocate appropriate resources to the athlete monitoring program that contribute to the research aims. These challenges have also been mitigated by using a phased approach for the research model implementation supported by (a) weekly meetings between researchers, athletic therapists, strength and conditioning coaches and the athletics department to obtain timely feedback on progress and obstacles; (b) utilization of freely available technologies for player tracking and monitoring (e.g., open-source form builders and analytics platforms) that did not depend on a large upfront financial investment; (c) project coordinators who work with the athletic therapy department to streamline preseason baseline testing of approximately 300 university athletes from 6 high-risk sports; (d) team representatives who coordinate in-season athlete monitoring and facilitate the development of feedback loops for athletes, coaches, and stakeholders; and (e) leveraging interdisciplinary collaborations within computer science and statistics to develop appropriate research methods and analytics to address the complexity of the research questions at hand. Although the training load-fatigue-injury research model proposed in this article requires careful planning and collaboration for implementation, it may better reflect the complexity of the sport injury problem faced by university and collegiate sport, hopefully leading to new insights on injury risk reduction methods that can increase athlete health and performance. This approach may also have broader applications outside of the university sport setting including for research and performance enhancement in related environments such as in professional sport, the military, and the workplace.

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References

1. Agel J, Rockwood T, Klossner D. Collegiate ACL injury rates across 15 sports: National Collegiate Athletic Association injury surveillance system data update (2004-2005 through 2012-2013). *Clin J Sport Med* 26: 518–523, 2016.
2. Alexe DI, Sandovici A, Robu V, et al. Measuring perceived social support in elite athletes: Psychometric properties of the Romanian version of the Multidimensional Scale of Perceived Social Support. *Percept Mot Skills* 128: 1197–1214, 2021.
3. Alt T, Breitenmoser T, Vonhoegen J, et al. The dynamic control ratio masks bilateral asymmetries–A gender-specific analysis of 264 healthy and ACL-injured athletes. *Res Sports Med* 30: 1–18, 2022.
4. Aquilina DA. Study of the relationship between elite athletes' educational development and sporting performance. *Int J Hist Sport* 30: 374–392, 2013.
5. Ardern CL, Taylor NF, Feller JA, Webster KE. Fear of re-injury in people who have returned to sport following anterior cruciate ligament reconstruction surgery. *J Sci Med Sport* 15: 488–495, 2012.
6. Beiter R, Nash R, McCrady M, et al. The prevalence and correlates of depression, anxiety, and stress in a sample of college students. *J Affect Disord* 173: 90–96, 2015.
7. Bender AM, Lawson D, Werthner P, Samuels CH. The clinical validation of the Athlete Sleep Screening Questionnaire: An instrument to identify athletes that need further sleep assessment. *Sports Med Open* 4: 23, 2018.
8. Benjaminse A, Webster KE, Kimp A, Meijer M, Gokeler A. Revised approach to the role of fatigue in anterior cruciate ligament injury prevention: A systematic review with meta-analyses. *Sports Med* 49: 565–586, 2019.
9. Bittencourt NFN, Meeuwisse WH, Mendonça LD, et al. Complex systems approach for sports injuries: Moving from risk factor identification to injury pattern recognition—Narrative review and new concept. *Br J Sports Med* 50: 1309–1314, 2016.
10. Bloom GA, Horton AS, McCrory P, Johnston KM. Sport psychology and concussion: New impacts to explore. *Br J Sports Med* 38: 519–521, 2004.
11. Bourne MN, Webster KE, Hewett TE. Is fatigue a risk factor for anterior cruciate ligament rupture? *Sports Med* 49: 1629–1635, 2019.
12. Brauer AA, Athey AB, Ross MJ, Grandner MA. Sleep and health among collegiate student athletes. *Chest* 156: 1234–1245, 2019.
13. Charest J, Grandner MA. Sleep and athletic performance: Impacts on physical performance, mental performance, injury risk and recovery, and mental health. *Sleep Med Clin* 15: 41–57, 2020.
14. Chin M, Leppänen M, Kulmala J-P, et al. A 3D motion capture analysis of a giving-way ankle episode during a 180-degree pivot turn: A case report. *J Biomech* 118: 110318, 2021.
15. Cinque ME, Godin JA, Moatshe G, et al. Do Tibial Plateau Fractures worsen outcomes of knee ligament injuries? A matched cohort analysis. *Orthop J Sports Med* 5: 2325967117723895, 2017.
16. Cohen S, Kamarck T, Mermelstein R. Perceived stress scale. *Meas Stress Guide Health Soc Sci* 10: 1–2, 1994.
17. Collins E, Doyle-Baker PK. Lesson learned when researching the lived experiences of young female skiers with knee injuries. *Med Sci Sports Exerc* 43: 360, 2011.
18. Coutts AJ, Reaburn P, Piva TJ, Rowsell GJ. Monitoring for overreaching in rugby league players. *Eur J Appl Physiol* 99: 313–324, 2007.

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19. Davis P, Halvarsson A, Lundström W, Lundqvist C. Alpine Ski Coaches' and Athletes' perceptions of factors influencing adaptation to stress in the classroom and on the slopes. *Front Psychol* 10: 1641, 2019.
20. Decker A, Aubertin P, Kriellaars D. Sleep and fatigue of elite circus student-artists during one year of training. *Med Probl Perform Art* 34: 125–131, 2019.
21. Delaney JS, Al-Kashmiri A, Correa JA. Mechanisms of injury for concussions in university football, ice hockey, and soccer. *Clin J Sport Med* 24: 233–237, 2014.
22. Dijkhuis TB, Blaauw FJ. Transferring targeted maximum likelihood estimation for causal inference into sports science. *Entropy* 24: 1060, 2022.
23. Driller MW, Dixon ZT, Clark MI. Accelerometer-based sleep behavior and activity levels in student athletes in comparison to student non-athletes. *Sport Sci Health* 13: 411–418, 2017.
24. Dudley RI, Lohman EB, Patterson CS, Knox KG, Gharibvand L. The relationship between kinesiophobia and biomechanics in anterior cruciate ligament reconstructed females. *Phys Ther Sport* 56: 32–37, 2022.
25. Emery CA, Pasanen K. Current trends in sport injury prevention. *Best Pract Res Clin Rheumatol* 33: 3–15, 2019.
26. Enoka RM, Duchateau J. Translating fatigue to human performance. *Med Sci Sports Exerc* 48: 2228–2238, 2016.
27. Escamilla RF, Barrentine SW, Fleisig GS, et al. Pitching biomechanics as a pitcher approaches muscular fatigue during a simulated baseball game. *Am J Sports Med* 35: 23–33, 2007.
28. Favero TG, White J. Periodization in college soccer. *Strength Cond J* 40: 33–44, 2018.
29. Fernandez WG, Yard EE, Comstock RD. Epidemiology of lower extremity injuries among U.S. High school athletes. *Acad Emerg Med* 14: 641–645, 2007.
30. Fields JB, Esco MR, Merrigan JJ, White JB, Jones MT. Internal training load measures during a competitive season in collegiate women lacrosse athletes. *Int J Exerc Sci* 13(4): 778–788, 2020.
31. Fogaca JL. Combining mental health and performance interventions: Coping and social support for student-athletes. *J Appl Sport Psychol* 33: 4–19, 2021.
32. Fonseca ST, Souza TR, Verhagen E, et al. Sports injury forecasting and complexity: A synergetic approach. *Sports Med* 50: 1757–1770, 2020.
33. Foster C, Rodriguez-Marroyo J, de Koning JJ. Monitoring training loads: The past, the present, and the future. *Int J Sports Physiol Perform* 12: S2–S8, 2017.
34. Fox A, Bonacci J, Hoffmann S, Nimphius S, Saunders N. Anterior cruciate ligament injuries in Australian football: Should women and girls be playing? You're asking the wrong question. *BMJ Open Sport Exerc Med* 6: e000778, 2020.
35. Fullagar HHK, Skorski S, Duffield R, et al. Sleep and athletic performance: The effects of sleep loss on exercise performance, and physiological and cognitive responses to exercise. *Sports Med* 45: 161–186, 2015.
36. Gabbett TJ, Hulin BT, Blanch P, Whiteley R. High training workloads alone do not cause sports injuries: How you get there is the real issue. *Br J Sports Med* 50: 444–445, 2016.
37. Gabbett TJ, Ullah S. Relationship between running loads and soft-tissue injury in elite team sport athletes. *J Strength Cond Res* 26: 953–960, 2012.
38. Gamble ASD, Mountjoy ML, Bigg JL, Spriet LL. Implementation of injury and illness surveillance protocols in varsity athletes. *Clin J Sport Med* 30: 321, 2020.
39. Gathercole R, Sporer B, Stellingwerff T, Sleivert G. Alternative countermovement-jump analysis to quantify acute neuromuscular fatigue. *Int J Sports Physiol Perform* 10: 84–92, 2015.
40. Gerometta A, Klouche S, Herman S, Lefevre N, Bohu Y. The Shoulder Instability-Return to Sport After Injury (SIRSI): A valid and reproducible scale to quantify psychological readiness to return to sport after traumatic shoulder instability. *Knee Surg Sports Traumatol Arthrosc* 26: 203–211, 2018.
41. Goerger BM, Marshall SW, Beutler AI, et al. Anterior cruciate ligament injury alters preinjury lower extremity biomechanics in the injured and uninjured leg: The JUMP-ACL study. *Br J Sports Med* 49: 188–195, 2015.
42. Govus AD, Coutts A, Duffield R, Murray A, Fullagar H. Relationship between pretraining subjective wellness measures, player load, and rating-of-perceived-exertion training load in American college football. *Int J Sports Physiol Perform* 13: 95–101, 2018.
43. Grandner MA, Hall C, Jaszewski A, et al. Mental health in student athletes: Associations with sleep duration, sleep quality, insomnia, fatigue, and sleep apnea symptoms. *Athl Train Sports Health Care* 13: e159–e167, 2021.
44. Grantham WJ, Byram IR, Meadows MC, Ahmad CS. The impact of fatigue on the kinematics of collegiate baseball pitchers. *Orthop J Sports Med* 2: 232596711453703, 2014.
45. Haddaway NR, Page MJ, Pritchard CC, McGuinness LA. PRISMA2020: An R package and Shiny App for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis. *Campbell Syst Rev* 18: e1230, 2022.
46. Halson SL. Monitoring training load to understand fatigue in athletes. *Sports Med* 44: 139–147, 2014.
47. Hamlin MJ, Wilkes D, Elliot CA, Lizamore CA, Kathiravel Y. Monitoring training loads and perceived stress in young elite university athletes. *Front Physiol* 10: 34, 2019.
48. Herman L, Foster C, Maher MA, Mikat RP, Porcari JP. Validity and reliability of the session RPE method for monitoring exercise training intensity. *South Afr J Sports Med* 18: 14–17, 2006.
49. Hooper SL, Mackinnon LT, Howard ALF, Gordon RD, Bachmann AW. Markers for monitoring overtraining and recovery. *Med Sci Sports Exerc* 27: 106–112, 1995.
50. Jeffries AC, Marcora SM, Coutts AJ, et al. Development of a revised conceptual framework of physical training for use in research and practice. *Sports Med* 52: 709–724, 2022.
51. Jordan MJ, Aagaard P, Herzog W. Asymmetry and thigh muscle coactivity in fatigued anterior cruciate ligament–reconstructed elite skiers. *Med Sci Sports Exerc* 49: 11–20, 2017.
52. Jordan MJ, Morris N, Barnert J, et al. Forecasting neuromuscular recovery after anterior cruciate ligament injury: Athlete recovery profiles with generalized additive modeling. *J Orthop Res*, 40: 2803–2812, 2022.
53. Kalkhoven JT, Watsford ML, Coutts AJ, Edwards WB, Impellizzeri FM. Training load and injury: Causal pathways and future directions. *Sports Med* 51: 1137–1150, 2021.
54. Kalkhoven JT, Watsford ML, Impellizzeri FM. A conceptual model and detailed framework for stress-related, strain-related, and overuse athletic injury. *J Sci Med Sport* 23: 726–734, 2020.
55. Kennedy SR, Dimick KM. Career maturity and professional sports expectations of college football and basketball players. *J Coll Stud Pers* 28: 293–297, 1987.
56. Kerbel Y, Smith C, Prodromo J, Nzeogu M, Mulcahey M. Epidemiology of hip and groin injuries in collegiate athletes in the United States. *Orthop J Sports Med* 6: 2325967118771676, 2018.
57. Kerr ZY, Evenson KR, Rosamond WD, et al. Association between concussion and mental health in former collegiate athletes. *Inj Epidemiol* 1: 1–10, 2014.
58. Kerr ZY, Marshall SW, Dompier TP, et al. College sports–related injuries—United States, 2009–10 through 2013–14 academic years. *Morb Mortal Wkly Rep* 64: 1330–1336, 2015.
59. Kiely J. Periodization theory: Confronting an inconvenient truth. *Sports Med* 48: 753–764, 2018.
60. Kluger B, Krupp L, Enoka R. Fatigue and fatigability in neurologic illnesses: Proposal for a unified taxonomy. *Neurology* 80: 409–416, 2013.
61. Kroshus E, Coakley S, Conway D, et al. Addressing mental health needs of NCAA student-athletes of colour: Foundational concepts from the NCAA summit on diverse student-athlete mental health and well-being. *Br J Sports Med* 57: 137–145, 2023.
62. Kroshus E, Garnett BR, Baugh CM, Calzo JP. Social norms theory and concussion education. *Health Educ Res* 30: 1004–1013, 2015.
63. Kroshus E, Wagner J, Wyrick D, et al. Wake up call for collegiate athlete sleep: Narrative review and consensus recommendations from the NCAA interassociation task force on sleep and wellness. *Br J Sports Med* 53: 731–736, 2019.
64. Li H, Moreland JJ, Peek-Asa C, Yang J. Preseason anxiety and depressive symptoms and prospective injury risk in collegiate athletes. *Am J Sports Med* 45: 2148–2155, 2017.
65. Lopes Dos Santos M, Uftring M, Stahl CA, et al. Stress in academic and athletic performance in collegiate athletes: A narrative review of sources and monitoring strategies. *Front Sports Act Living* 2: 42, 2020.
66. Lucarno S, Zago M, Buckthorpe M, et al. Systematic video analysis of anterior cruciate ligament injuries in professional female soccer players. *Am J Sports Med* 49: 1794–1802, 2021.
67. Mah CD, Kezirian EJ, Marcello BM, Dement WC. Poor sleep quality and insufficient sleep of a collegiate student-athlete population. *Sleep Health* 4: 251–257, 2018.
68. Mann JB, Bryant KR, Johnstone B, Ivey PA, Sayers SP. Effect of physical and academic stress on illness and injury in division 1 college football players. *J Strength Cond Res* 30: 20–25, 2016.

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69. McKay AKA, Stellingwerff T, Smith ES, et al. Defining training and performance caliber: A participant classification framework. *Int J Sports Physiol Perform* 17: 317–331, 2021.
70. McLaren SJ, Macpherson TW, Coutts AJ, et al. The relationships between internal and external measures of training load and intensity in team sports: A meta-analysis. *Sports Med* 48: 641–658, 2018.
71. Meeuwisse WH. Assessing causation in sport injury: A multifactorial model. *Clin J Sport Med* 4: 166–170, 1994.
72. Meeuwisse WH, Sellmer R, Hagel BE. Rates and risks of injury during intercollegiate basketball. *Am J Sports Med* 31: 379–385, 2003.
73. Meeuwisse WH, Tyreman H, Hagel B, Emery C. A dynamic model of etiology in sport injury: The recursive nature of risk and causation. *Clin J Sport Med* 17: 215–219, 2007.
74. Monma T, Ando A, Asanuma T, et al. Sleep disorder risk factors among student athletes. *Sleep Med* 44: 76–81, 2018.
75. Moreland JJ, Coxe KA, Yang J. Collegiate athletes' mental health services utilization: A systematic review of conceptualizations, operationalizations, facilitators, and barriers. *J Sport Health Sci* 7: 58–69, 2018.
76. Murray A, Buttfield A, Simpkin A, Sproule J, Turner AP. Variability of within-step acceleration and daily wellness monitoring in Collegiate American Football. *J Sci Med Sport* 22(4): 488–493, 2019.
77. National Institutes of Health and National Heart, Lung, and Blood Institute. *Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies*. 2014.
78. Nicholls AR, Levy AR, Perry JL. Emotional maturity, dispositional coping, and coping effectiveness among adolescent athletes. *Psychol Sport Exerc* 17: 32–39, 2015.
79. Nielsen RO, Bertelsen ML, Møller M, et al. Training load and structure-specific load: Applications for sport injury causality and data analyses. *Br J Sports Med* 52: 1016–1017, 2018.
80. Olsen O-E, Myklebust G, Engebretsen L, Bahr R. Injury mechanisms for anterior cruciate ligament injuries in team handball: A systematic video analysis. *Am J Sports Med* 32: 1002–1012, 2004.
81. Paquette M, Melcher DA. Impact of a long run on injury-related biomechanics with relation to weekly mileage in trained male runners. *J Appl Biomech* 33: 216–221, 2017.
82. Parsons JL, Coen SE, Bekker S. Anterior cruciate ligament injury: Towards a gendered environmental approach. *Br J Sports Med* 55: 984–990, 2021.
83. Passfield L, Murias JM, Sacchetti M, Nicolò A. Validity of the training-load concept. *Int J Sports Physiol Perform* 17: 507–514, 2022.
84. Raikes AC, Athey A, Alfonso-Miller P, Killgore WDS, Grandner MA. Insomnia and daytime sleepiness: Risk factors for sports-related concussion. *Sleep Med* 58: 66–74, 2019.
85. Register-Mihalik JK, Marshall SW, Kay MC, et al. Perceived social norms and concussion-disclosure behaviours among first-year NCAA student-athletes: Implications for concussion prevention and education. *Res Sports Med* 29: 1–11, 2021.
86. Roy X, Comtois AS, Sercia P. Use of the session-RPE method to quantify in-season training loads in Canadian University football players. *J Aust Strength Cond* 28: 47–56, 2020.
87. Sampson JA, Murray A, Williams S, Sullivan A, Fullagar HHK. Subjective wellness, acute: Chronic workloads, and injury risk in college football. *J Strength Cond Res* 33: 3367–3373, 2019.
88. Samuels C, James L, Lawson D, Meeuwisse W. The athlete sleep screening questionnaire: A new tool for assessing and managing sleep in elite athletes. *Br J Sports Med* 50: 418–422, 2016.
89. Schick DM, Meeuwisse WH. Injury rates and profiles in female ice hockey players. *Am J Sports Med* 31: 47–52, 2003.
90. Scott TJ, Black CR, Quinn J, Coutts AJ. Validity and reliability of the session-RPE method for quantifying training in Australian football: A comparison of the CR10 and CR100 scales. *J Strength Cond Res* 27: 270–276, 2013.
91. Senevirathna AM, Pohl AJ, Jordan MJ, Edwards WB, Ferber R. Differences in kinetic variables between injured and uninjured rearfoot runners: A hierarchical cluster analysis. *Scand J Med Sci Sports*, 33:160–168, 2023.
92. Silvia A, Narciso FV, Soalheiro I, et al. Poor sleep quality's association with soccer injuries: Preliminary data. *Int J Sports Physiol Perform* 15: 671–676, 2019.
93. Smith AM. Psychological impact of injuries in athletes. *Sports Med* 22: 391–405, 1996.
94. Stern BD, Hegedus EJ, Lai Y-C. Injury prediction as a non-linear system. *Phys Ther Sport* 41: 43–48, 2020.
95. Sullivan M, Moore M, Blom LC, Slater G. Relationship between social support and depressive symptoms in collegiate student athletes. *J Study Sports Athletes Educ* 14: 192–209, 2020.
96. Sutcliffe JH, Greenberger PA. Identifying psychological difficulties in college athletes. *J Allergy Clin Immunol Pract* 8: 2216–2219, 2020.
97. Thomas AC, Lepley LK, Wojtys EM, McLean SG, Palmieri-Smith RM. Effects of neuromuscular fatigue on quadriceps strength and activation and knee biomechanics in individuals post–anterior cruciate ligament reconstruction and healthy adults. *J Orthop Sports Phys Ther* 45: 1042–1050, 2015.
98. Thomas AC, McLean SG, Palmieri-Smith RM. Quadriceps and hamstrings fatigue alters hip and knee mechanics. *J Appl Biomech* 26: 159–170, 2010.
99. Thorpe RT, Atkinson G, Drust B, Gregson W. Monitoring fatigue status in elite team sport athletes: Implications for practice. *Int J Sports Physiol Perform* 12(Supp 2): S227–S234, 2017.
100. Udry E, Gould D, Bridges D, Beck L. Down but not out: Athlete responses to season-ending injuries. *J Sport Exerc Psychol* 19: 229–248, 1997.
101. Van Mechelen W, Hlobil H, Kemper HC. Incidence, severity, aetiology and prevention of sports injuries. *Sports Med* 14: 82–99, 1992.
102. Venter RE. Role of sleep in performance and recovery of athletes: A review article. *South Afr J Res Sport Phys Educ Recreat* 34: 167–184, 2012.
103. Verheul J, Nedergaard NJ, Vanrenterghem J, Robinson MA. Measuring biomechanical loads in team sports–from lab to field. *Sci Med Footb* 4: 246–252, 2020.
104. von Rosen P, Frohm A, Kottorp A, Fridén C, Heijne A. Too little sleep and an unhealthy diet could increase the risk of sustaining a new injury in adolescent elite athletes. *Scand J Med Sci Sports* 27: 1364–1371, 2017.
105. Wade KH, Richmond RC, Smith GD. Physical activity and longevity: How to move closer to causal inference. *Br J Sports Med* 52: 890–891, 2018.
106. Wall BT, Morton JP, van Loon LJC. Strategies to maintain skeletal muscle mass in the injured athlete: Nutritional considerations and exercise mimetics. *Eur J Sport Sci* 15: 53–62, 2015.
107. Watson A, Johnson M, Sanfilippo J. Decreased sleep is an independent predictor of in-season injury in male collegiate basketball players. *Orthop J Sports Med* 8: 2325967120964481, 2020.
108. Wilkerson GB, Colston MA, Baker CS. A sport fitness index for assessment of sport-related injury risk. *Clin J Sport Med* 26: 423–428, 2016.
109. Wilkerson GB, Gupta A, Allen JR, Keith CM, Colston MA. Utilization of practice session average inertial load to quantify college football injury risk. *J Strength Cond Res* 30: 2369–2374, 2016.
110. Williams JH, Hoffman S, Jaskowak DJ, Tegarden D. Physical demands and physiological responses of extra time matches in collegiate women's soccer. *Sci Med Footb* 3: 307–312, 2019.
111. Williams KR. Biomechanical factors contributing to marathon race success. *Sports Med* 37: 420–423, 2007.
112. Williams S, West S, Cross M, Stokes K. Better way to determine the acute: Chronic workload ratio? *Br J Sports Med* 51: 209–210, 2017.
113. Windt J, Gabbett T. How do training and competition workloads relate to injury? The workload—injury aetiology model. *Br J Sports Med* 51: 428–435, 2017.
114. Wojtys EM, Beaulieu ML, Ashton-Miller J. New perspectives on ACL injury: On the role of repetitive sub-maximal knee loading in causing ACL fatigue failure. *J Orthop Res* 34: 2059–2068, 2016.
115. Zago M, David S, Bertozzi F, et al. Fatigue induced by repeated changes of direction in élite female football (soccer) players: Impact on lower limb biomechanics and implications for ACL injury prevention. *Front Bioeng Biotechnol* 9: 586, 2021.
116. Zaryski C, Smith DJ. Training principles and issues for ultra-endurance athletes. *Curr Sports Med Rep* 4: 165–170, 2005.
117. Zimet GD, Dahlem NW, Zimet SG, Farley GK. The multidimensional scale of perceived social support. *J Pers Assess* 52: 30–41, 1988.
118. Zuckerman SL, Totten DJ, Rubel KE, et al. Mechanisms of injury as a diagnostic predictor of sport-related concussion severity in football, basketball, and soccer: Results from a regional concussion registry. *Neurosurgery* 63: 102–112, 2016.

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